

No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Intelligent Crop Recommendation	Predicts the most suitable crop using soil nutrients (N, P, K) and environmental conditions such as temperature, humidity, pH, and rainfall.	ML Prediction Engine	Done	Core Feature
2	Soil & Climate Data Input	Accepts user input through an intuitive web interface for accurate crop prediction.	User Interface	Done	High
3	Machine Learning Prediction Model	Uses the trained Random Forest model to generate highly accurate crop recommendations.	Machine Learning	Done	Core Feature
4	Prediction Confidence Score	Displays confidence percentage along with the recommended crop to improve user trust.	Prediction Module	Done	Medium
5	Agricultural Data Visualization	Presents crop distribution, soil analysis, and model insights using interactive charts and graphs.	Analytics Dashboard	Done	High
6	Dataset Preprocessing	Cleans, validates, and transforms agricultural data before model training and prediction.	Data Processing	Done	High

7	Model Performance Evaluation	Evaluates the model using Accuracy, Confusion Matrix, and Classification Report for reliable predictions.	Evaluation Module	Done	Medium
8	Responsive Web Application	Provides an easy-to-use interface accessible from desktop and mobile browsers.	Frontend (Flask UI)	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (Crop Recommendation, ML Prediction, Data Input, Dataset Processing, Model Evaluation)
Additional / Nice-to-Have Features	Agricultural Insights, Future Feature Support
Features Tested & Verified	Soil Data Input, Crop Recommendation, Crop Suitability Analysis, ML Prediction, Prediction Dashboard

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Soil data input form, Prediction result dashboard
2	Backend / Logic	3	Crop recommendation, Crop suitability analysis, Input validation
3	Database / Storage	1	Crop dataset (CSV) and trained ML model
4	API / Integration	1	Flask application connecting frontend with the prediction model
5	Security / Authentication	1	Input Validation and Secure Request Handling